

Physical Chemistry (I)

Lecture 20

Colligative Properties
Chemical Reactions



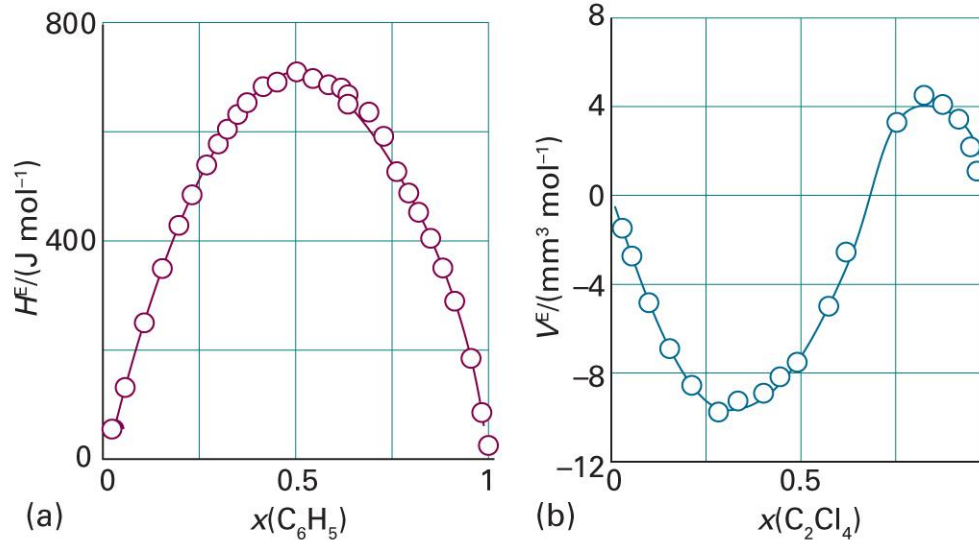
In the Last Class

- Deviations from ideality
 - Excess function
 - Activity
- Phase diagram of mixture
 - Ideal solutions
 - Real solutions: azeotropes
 - Real solutions: phase separation

Excess Function

$$X^E = \Delta_{\text{mix}}X - \Delta_{\text{mix}}X^{\text{ideal}}$$

- Regular solution: $S^E = 0$ and $U^E \neq 0$ (or $H^E \neq 0$)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.3)

Activity a_J

- Ideal-dilute solution

- Solvent: $\mu_A(l) = \mu_A^*(l) + RT \ln x_A$

- Solute: $\mu_B(l) = \mu_B^\ominus(l) + RT \ln x_B$

- where $\mu_B^\ominus(l) = \mu_B^*(l) + RT \ln(K_B/p_B^*)$

- Real solution

- Solvent: $\mu_A(l) = \mu_A^*(l) + RT \ln a_A$

- Solute: $\mu_B(l) = \mu_B^\ominus(l) + RT \ln a_B$

Last class

BW Model: Margules Equations

$$\gamma_A = e^{\xi x_B^2}, \quad \gamma_B = e^{\xi x_A^2}$$

$$p_A = p_A^* a_A = p_A^* x_A e^{\xi(1-x_A)^2}$$

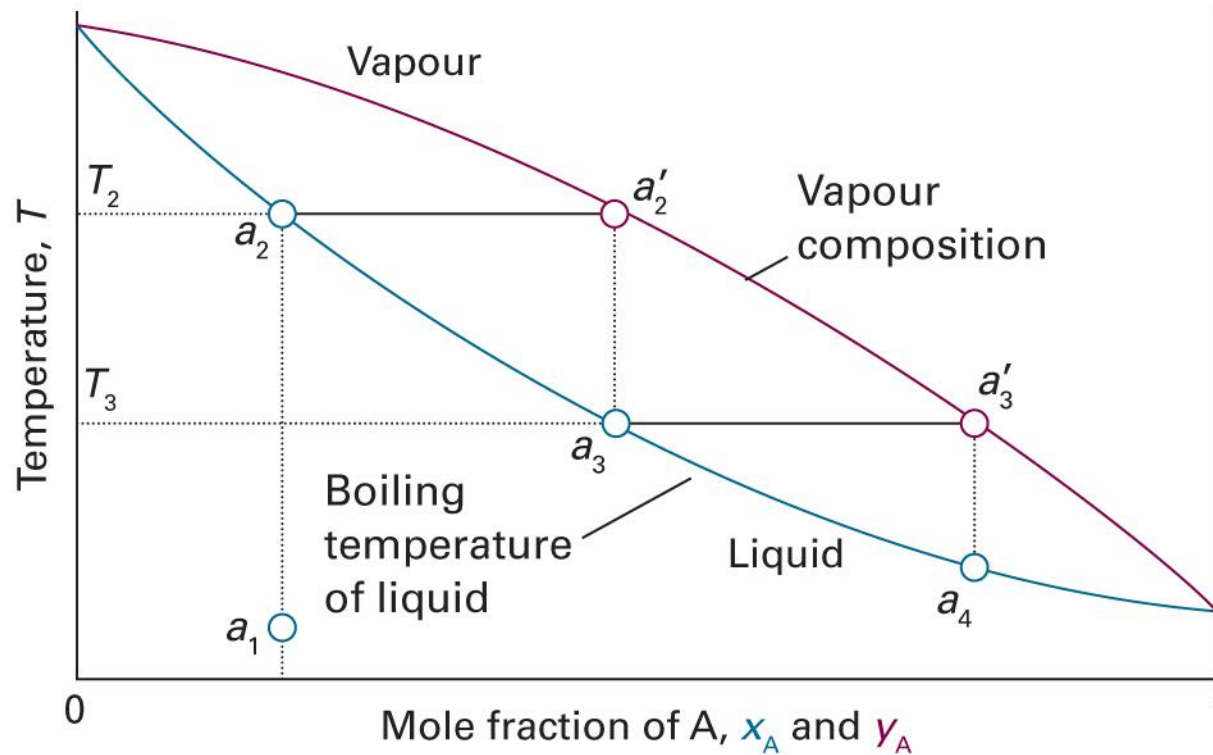
- $x_A \rightarrow 1$: $p_A = p_A^* x_A e^0 = p_A^* x_A$ (Raoult's law)
- $x_A \rightarrow 0$: $p_A = p_A^* x_A e^{\xi} = K_A x_A$ (Henry's law)

$$\text{where } K_A = p_A^* e^{\xi}$$

Last class

Phase Diagram: Ideal Solutions

$$p_A^* > p_B^* \text{ (A is more volatile)}$$

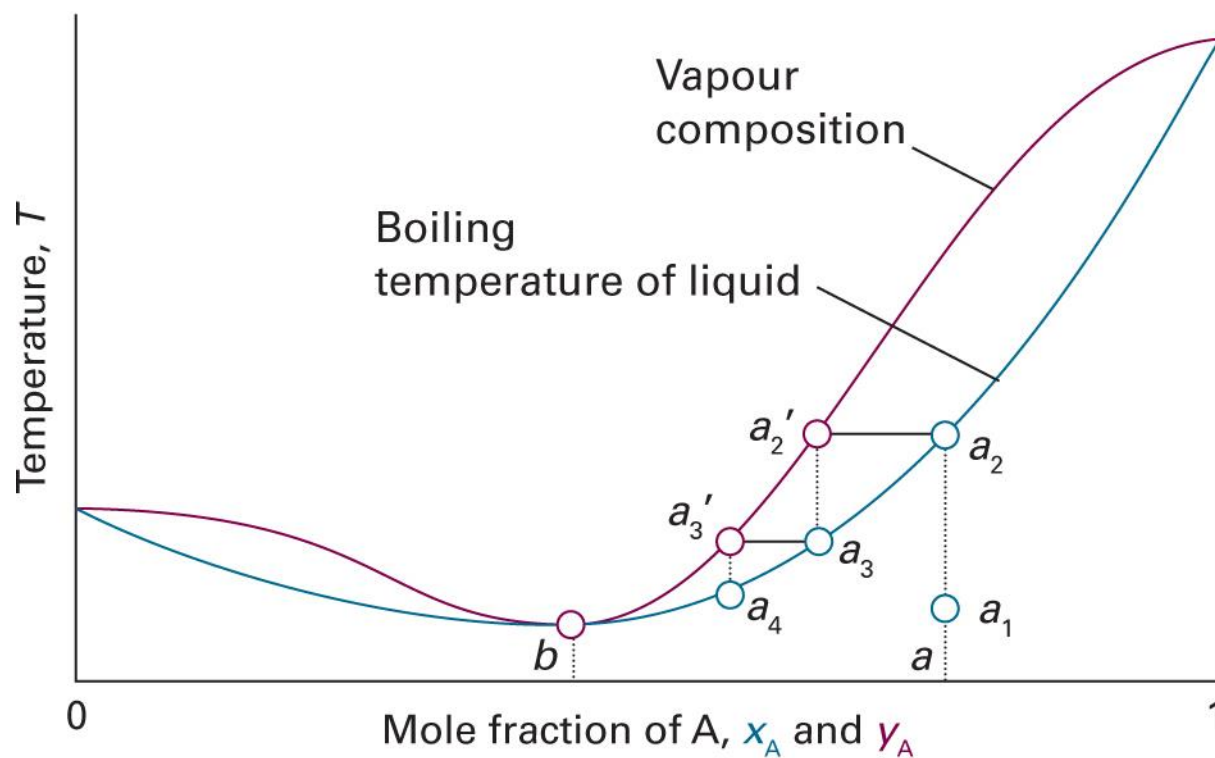


(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.4)



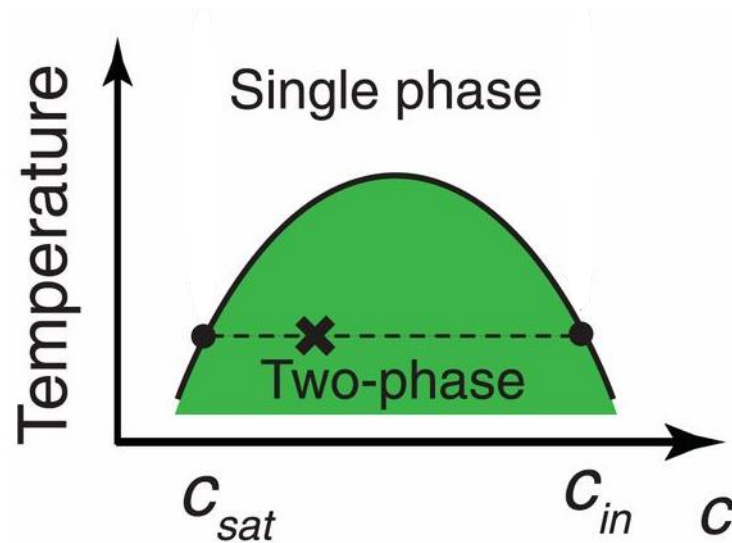
Low-Boiling Azeotrope

$G^E > 0$ (A-B interactions weaker)



Last class

Phase Separation



Lever rule applies in the two-phase region!

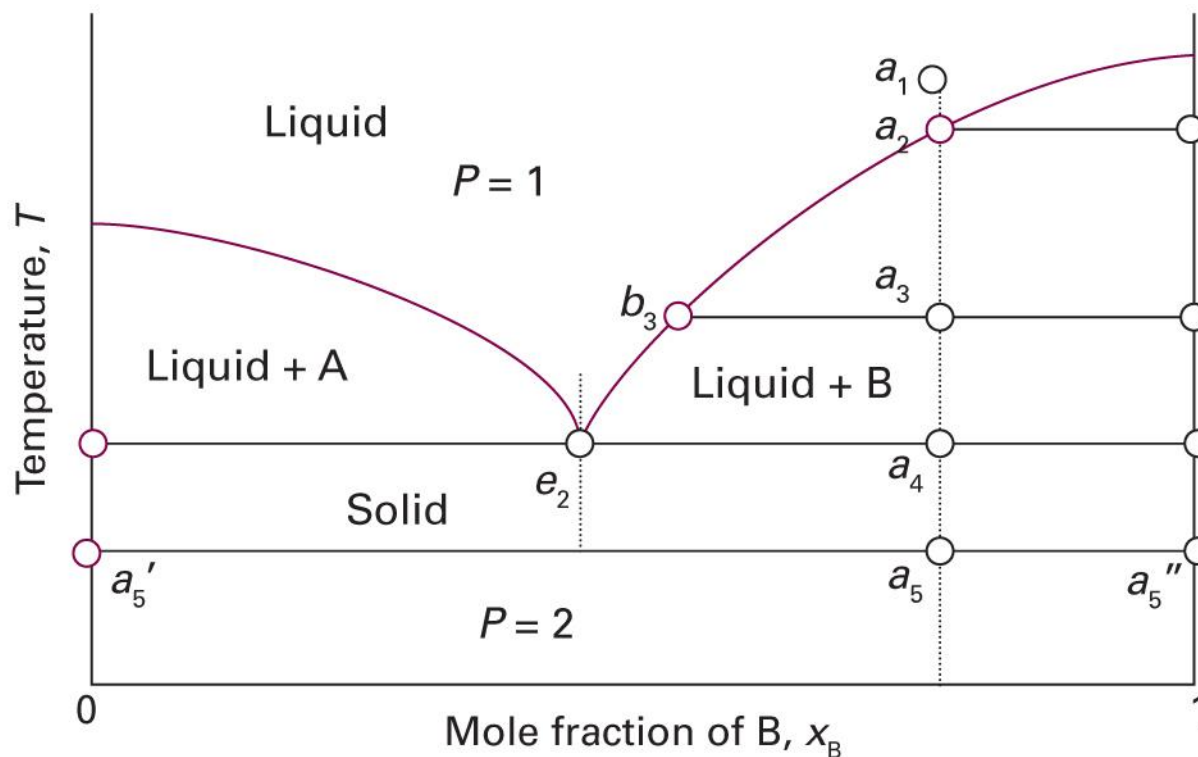
Liquid-Solid Equilibrium

Consider a two-component system (A and B)
where the two solids are almost immiscible.

Upon mixing,

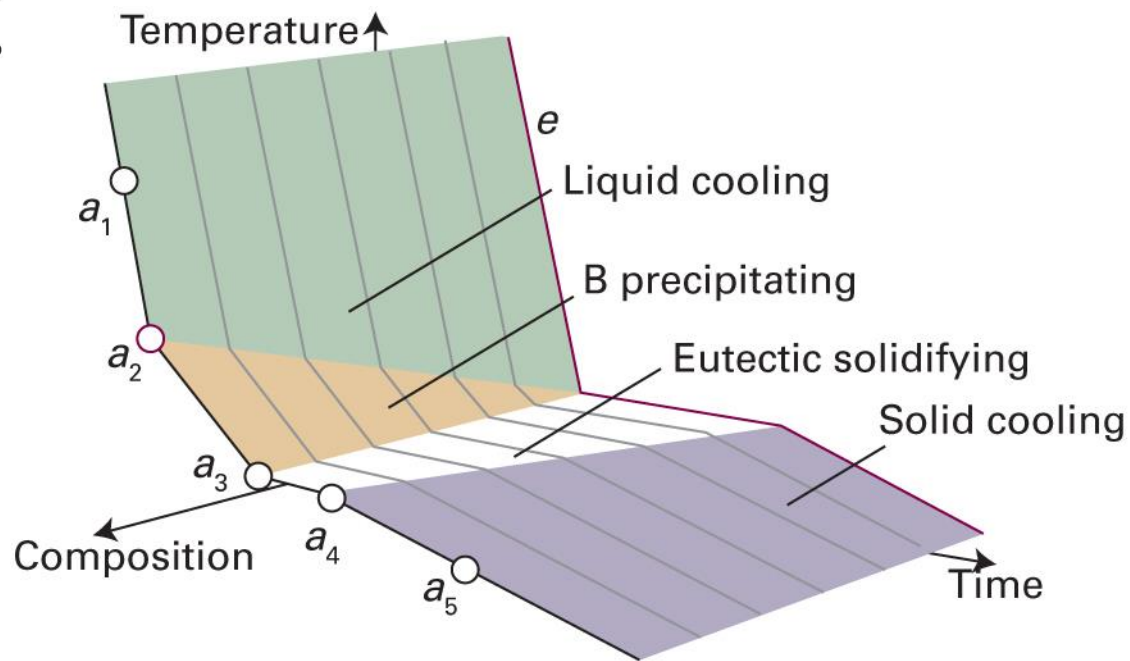
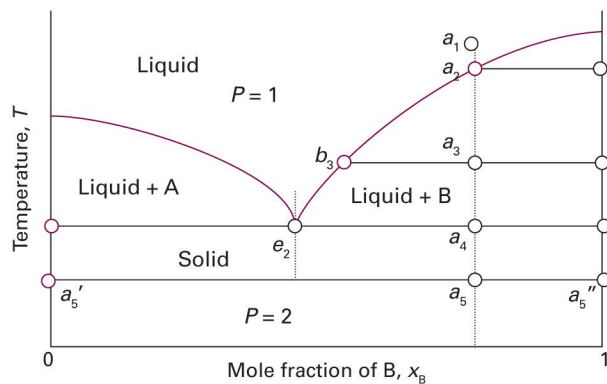
- Melting temperature ↓: A-B interactions are weaker
- Melting temperature ↑: A-B interactions are stronger

Phase Diagrams Involving Solids



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5D.1)

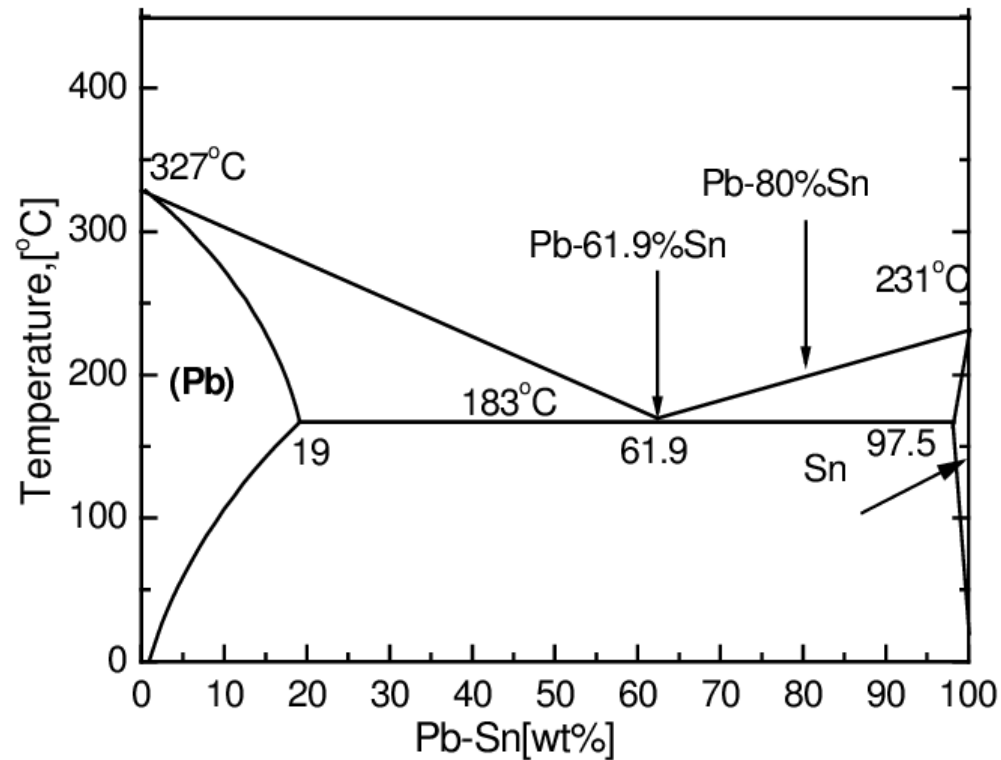
Eutectics



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 5D.1 and 5D.2)

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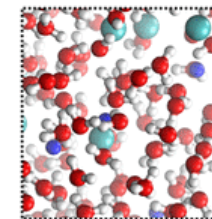
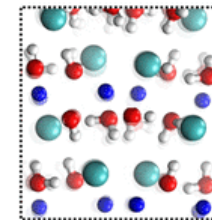
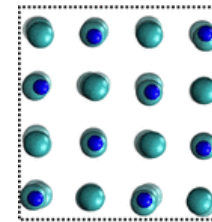
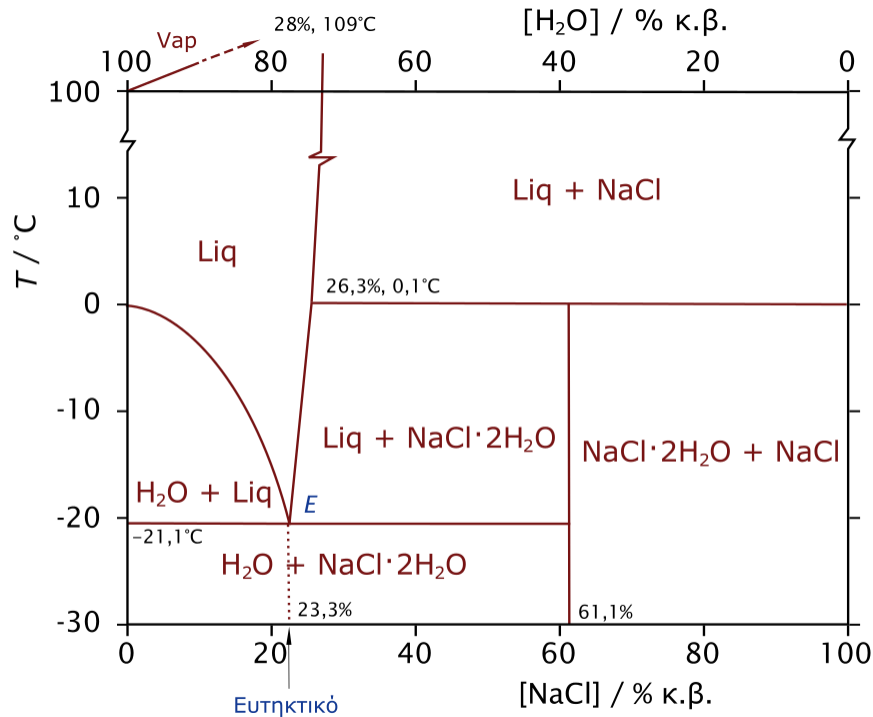
Pb-Sn System



(Image: https://www.researchgate.net/figure/The-Pb-Sn-phase-diagram_fig1_291162603)

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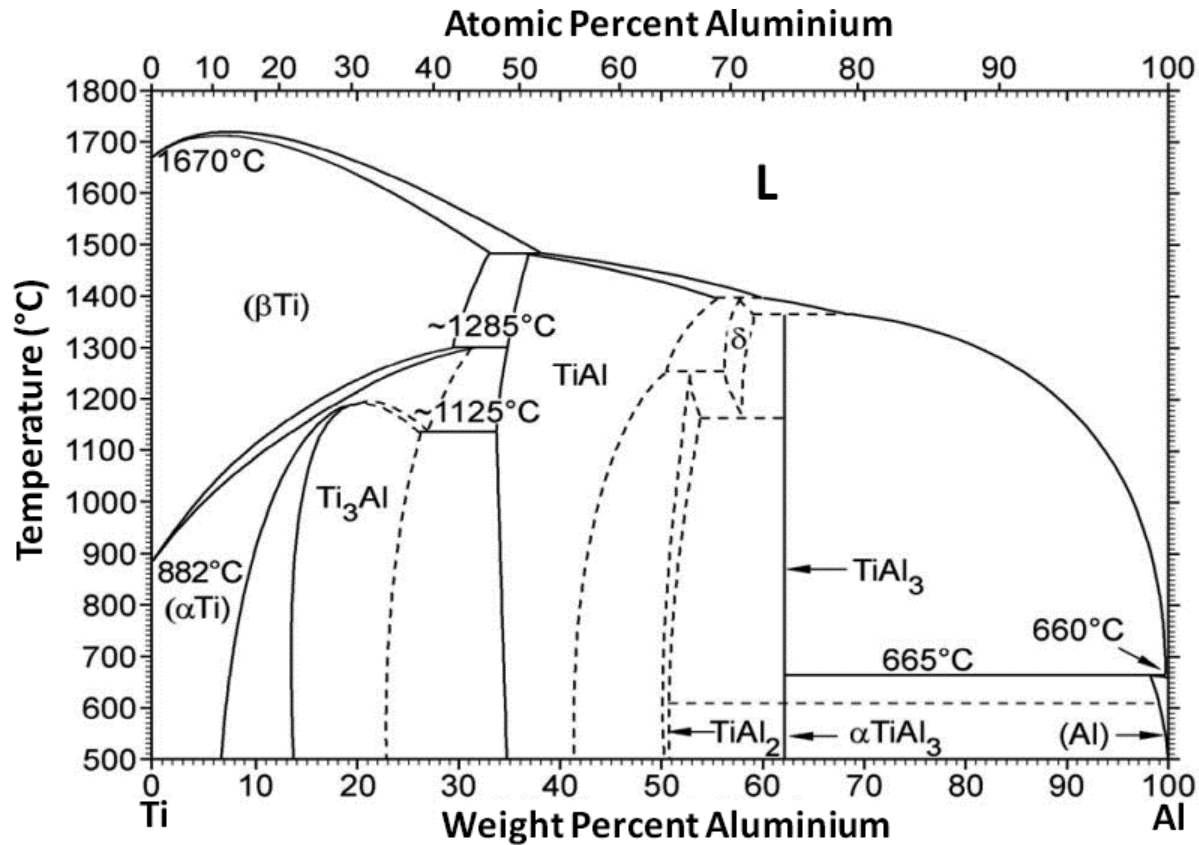
H₂O-NaCl System



(Image: <https://commons.wikimedia.org/wiki/File:H2O-NaCl-phase-diagram-greek.svg>)

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Ti-Al System



(Image: Mitra and Wanhill, *Structural Intermetallics* (2016), Figure 10)

Liquid-Solid Solutions

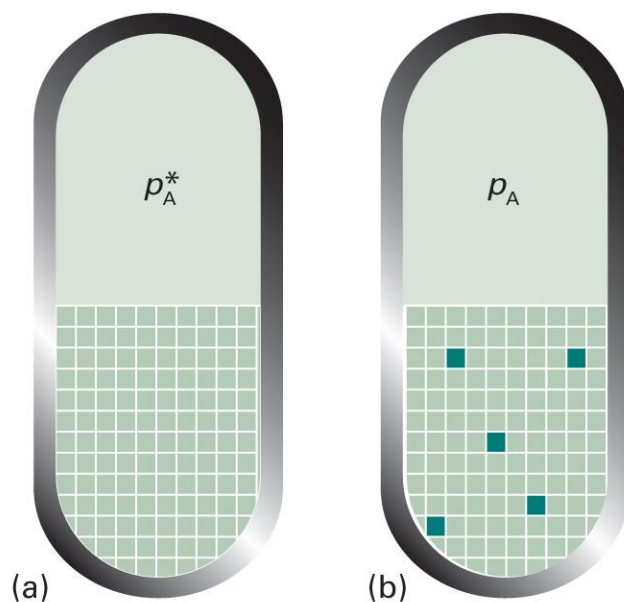
- Considered as ideal-dilute solutions
 - Liquid: solvent
 - Solid: solute
- Two assumptions
 - The solute is not volatile
 - The solute does not dissolve in the solid solvent

Colligative Properties

- Physical properties that depends on the relative number of solute particles present but not their chemical identity
1. Lowering of vapor pressure
 2. Elevation of boiling point
 3. Depression of freezing point
 4. Osmotic pressure

Origin of Colligative Properties

- Due to the **reduction of the chemical potential** of the liquid solvent as a result of the presence of solute



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.7)

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Lowering of Vapor Pressure

From Raoult's law,

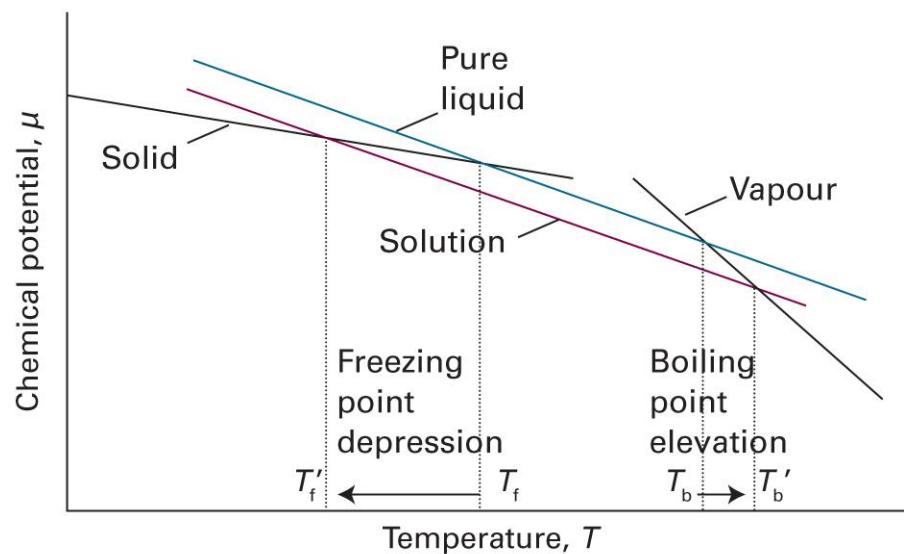
$$p_A = x_A p_A^*$$

The change in vapor pressure is

$$\Delta p_A = p_A^* - p_A = (1 - x_A) p_A^* = x_B p_A^*$$

Changes in Phase Transitions

$$\mu_A = \mu_A^* + RT \ln x_A \quad (x_A < 1)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.6)

Boiling: Liquid-Gas Equilibrium

$$\mu_A^*(g) = \mu_A(l)$$

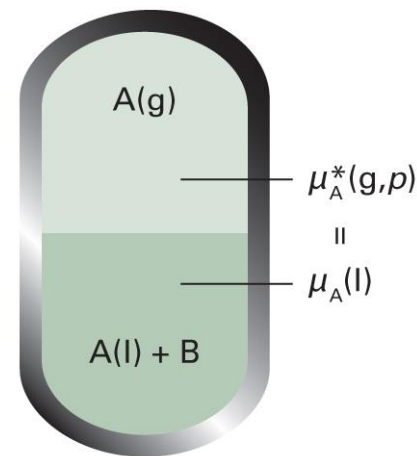
$$\mu_A^*(g) = \mu_A^*(l) + RT \ln x_A$$

which can be rearranged as

$$\ln x_A = \frac{\mu_A^*(g) - \mu_A^*(l)}{RT} = \frac{\Delta_{\text{vap}}G}{RT}$$

Using the Gibbs-Helmholtz equation,

$$\frac{d \ln x_A}{dT} = \frac{1}{R} \frac{d(\Delta_{\text{vap}}G/T)}{dT} = -\frac{\Delta_{\text{vap}}H}{RT^2}$$



Integration

$$\frac{d \ln x_A}{dT} = - \frac{\Delta_{\text{vap}} H}{RT^2}$$

$$d \ln x_A = - \frac{\Delta_{\text{vap}} H}{RT^2} dT$$

$$\int_{\text{pure solvent}}^{\text{current state}} d \ln x_A = - \frac{1}{R} \int_{\text{pure solvent}}^{\text{current state}} \frac{\Delta_{\text{vap}} H}{T^2} dT$$

$$\int_0^{\ln x_A} d \ln x'_A = - \frac{1}{R} \int_{T^*}^T \frac{\Delta_{\text{vap}} H}{(T')^2} dT'$$

Evaluation

$$\int_0^{\ln x_A} d \ln x'_A = -\frac{1}{R} \int_{T^*}^T \frac{\Delta_{\text{vap}} H}{(T')^2} dT'$$

Assuming that $\Delta_{\text{vap}} H$ is constant,

$$\ln x_A = -\frac{\Delta_{\text{vap}} H}{R} \int_{T^*}^T \frac{1}{(T')^2} dT' = -\frac{\Delta_{\text{vap}} H}{R} \left(-\frac{1}{T} + \frac{1}{T^*} \right)$$

$$\ln(1 - x_B) = \frac{\Delta_{\text{vap}} H}{R} \left(\frac{1}{T} - \frac{1}{T^*} \right)$$

Two (Valid) Approximations

$$\ln(1 - x_B) = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T} - \frac{1}{T^*} \right)$$

For $x_B \ll 1$, $\ln(1 - x_B) \approx -x_B$ and

$$x_B \approx \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T^*} - \frac{1}{T} \right) = \frac{\Delta_{\text{vap}}H}{RTT^*} (T - T^*)$$

Also, supposing $\Delta T_b = T - T^* \ll T^*$,

$$x_B \approx \frac{\Delta_{\text{vap}}H}{R(T^*)^2} \Delta T_b$$

Elevation of Boiling Point

$$x_B \approx \frac{\Delta_{\text{vap}}H}{R(T^*)^2} \Delta T_b$$

Rearrangement gives

$$\Delta T_b = K x_B$$

where $K = R(T^*)^2 / \Delta_{\text{vap}}H$.

Using the molality,

$$\Delta T_b = K_b b$$

where K_b is called the boiling-point constant.

Freezing: Solid-Liquid Equilibrium

$$\mu_A^*(s) = \mu_A(l)$$

$$\mu_A^*(s) = \mu_A^*(l) + RT \ln x_A$$

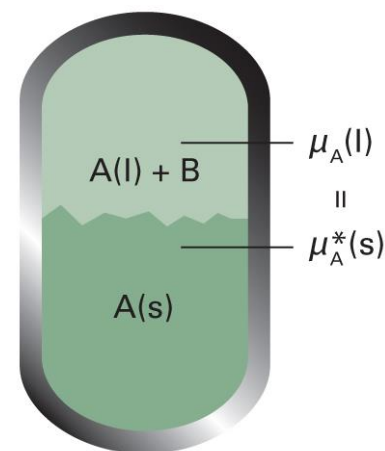
Similarly, we obtain

$$\Delta T_f = K' x_B$$

where $K' = R(T^*)^2 / \Delta_{\text{fus}}H$, or,

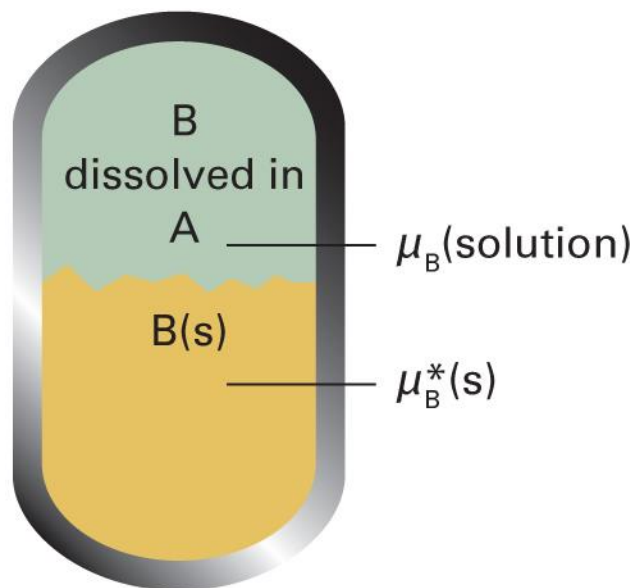
$$\Delta T_f = K_f b$$

where K_f is called the freezing-point constant.



Solubility

- **Saturation:** the undissolved solute in equilibrium with the dissolved solute



Chemical Potential of Solute

For solute B,

$$\mu_B^*(s) = \mu_B(l) = \mu_B^*(l) + RT \ln x_B$$

Rearrangement leads to

$$\ln x_B = \frac{\mu_B^*(s) - \mu_B^*(l)}{RT} = -\frac{\Delta_{\text{fus}}G}{RT}$$

Differentiating both sides and integrating,

$$\int_0^{\ln x_B} d \ln x'_B = \frac{1}{R} \int_{T_f}^T \frac{\Delta_{\text{fus}}H}{(T')^2} dT'$$

Solubility

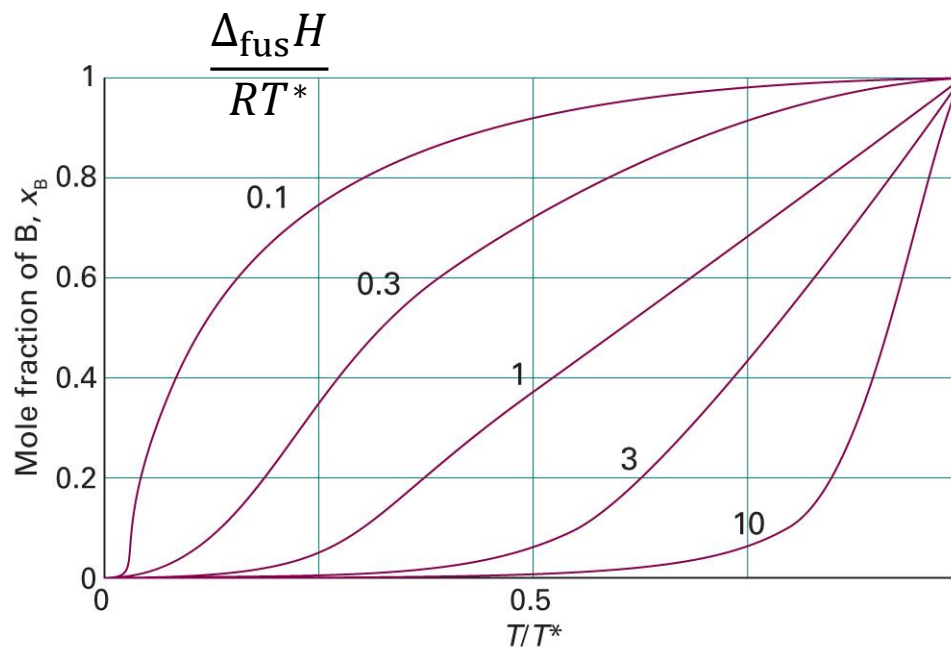
$$\int_0^{\ln x_B} d \ln x'_B = \frac{1}{R} \int_{T_f}^T \frac{\Delta_{\text{fus}} H}{(T')^2} dT'$$

Assuming that $\Delta_{\text{fus}} H$ is constant,

$$\ln x_B = \frac{\Delta_{\text{fus}} H}{R} \left(\frac{1}{T_f} - \frac{1}{T} \right)$$

(Idealized) Solubility

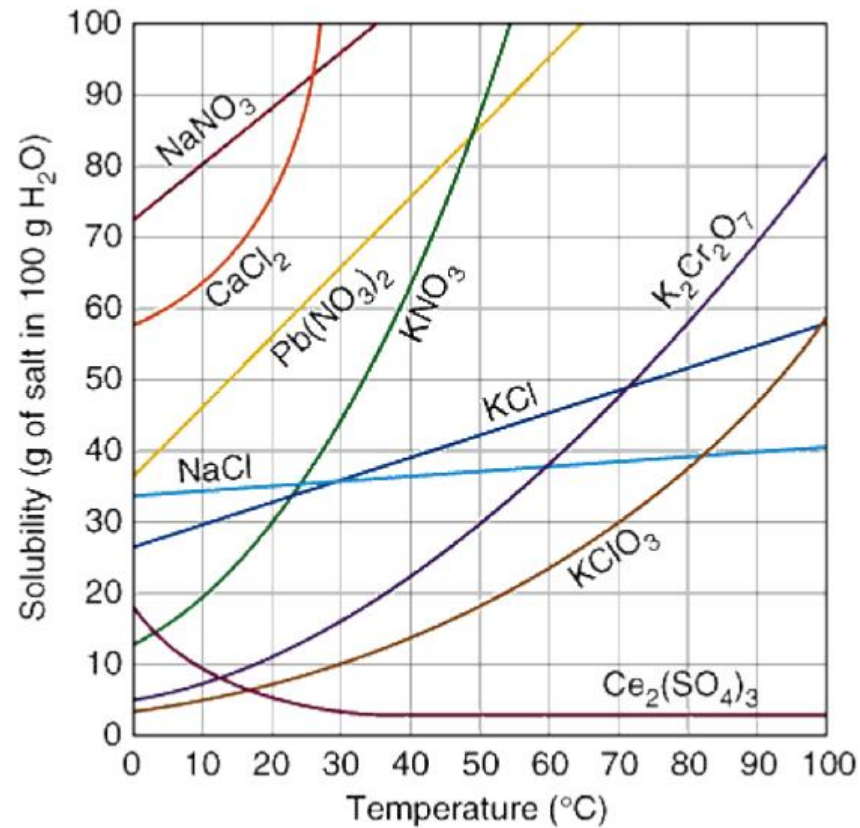
$$\ln x_B = \frac{\Delta_{\text{fus}}H}{R} \left(\frac{1}{T^*} - \frac{1}{T} \right)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.11)

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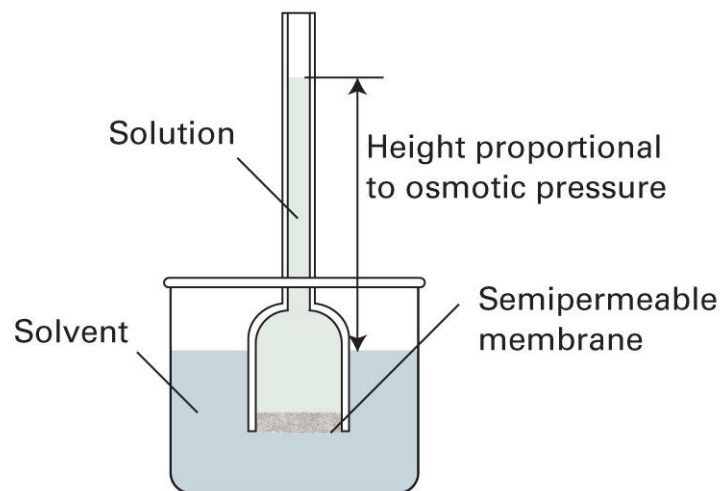
Real Solubility Data



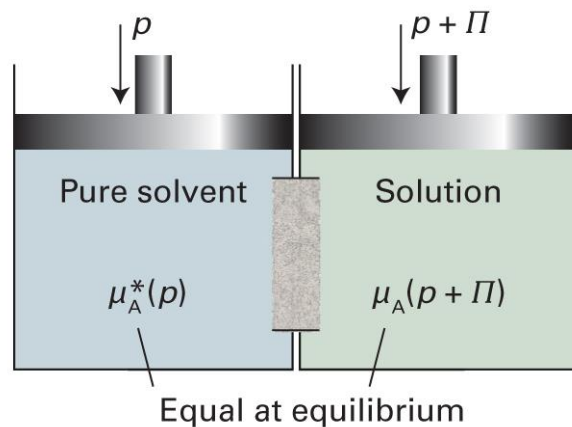
(Image: <https://socratic.org/questions/what-are-solubility-curves>)

Osmosis

the spontaneous passage of a pure solvent into a solution separated from it by a semipermeable membrane



Thermodynamics of Osmosis



$$\mu_A^*(p) = \mu_A(x_A, p + \Pi)$$

$$\mu_A^*(p) = \mu_A^*(p + \Pi) + RT \ln x_A$$

Thus,

$$\mu_A^*(p + \Pi) = \mu_A^*(p) - RT \ln x_A$$

Pressure Effect

In general,

$$G_m(p_2) = G_m(p_1) + \int_{p_1}^{p_2} V_m dp$$

$$\mu_A^*(p + \Pi) = \mu_A^*(p) + \int_p^{p+\Pi} V_m dp'$$

Assuming that V_m is constant,

$$\mu_A^*(p + \Pi) = \mu_A^*(p) + V_m \int_p^{p+\Pi} 1 dp' = \mu_A^*(p) + V_m \Pi$$

Combining the Two

$$\mu_A^*(p + \Pi) = \mu_A^*(p) - RT \ln x_A$$

$$\mu_A^*(p + \Pi) = \mu_A^*(p) + V_m \Pi$$

Comparing the two equations,

$$-RT \ln x_A = V_m \Pi$$

$$-RT \ln(1 - x_B) = V_m \Pi$$

Since $\ln(1 - x_B) \approx -x_B$ for $x_B \ll 1$,

$$RT x_B = V_m \Pi$$

Osmotic Pressure

$$RTx_B = V_m\Pi$$

For a dilute solution,

$$x_B = n_B/(n_A + n_B) \approx n_B/n_A$$

Thus,

$$RTn_B = n_A V_m \Pi = V \Pi$$

and

$$\Pi = [B]RT \text{ (van't Hoff equation)}$$

Real Solutions

For an ideal solution (now solute = J),

$$\Pi = [J]RT$$

The osmotic virial expansion is

$$\Pi = [J]RT(1 + B[J] + \dots)$$

where B is called the **osmotic virial coefficient**.

Osmometry

$$\Pi = [J]RT(1 + B[J] + \dots)$$

Taking the first correction term only,

$$\Pi \approx [J]RT(1 + B[J])$$

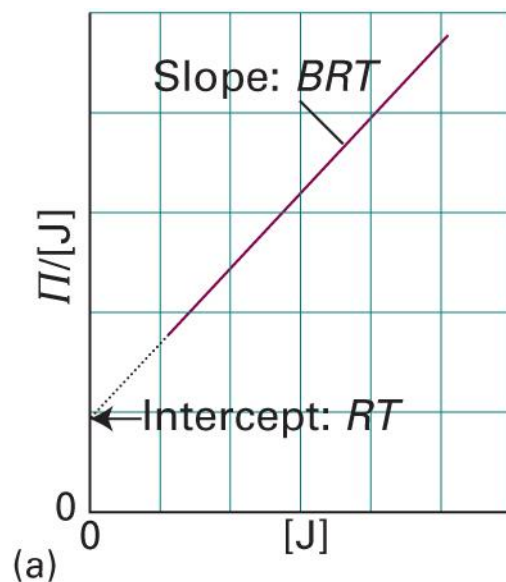
$$\frac{\Pi}{[J]} \approx RT + BRT[J]$$

The mass concentration is $c_{\text{mass},J} = [J]M$;

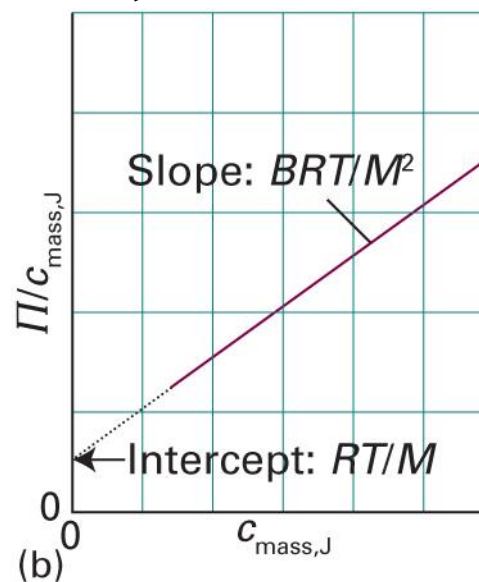
$$\frac{\Pi}{c_{\text{mass},J}} \approx \frac{RT}{M} + \frac{BRT}{M^2} c_{\text{mass},J}$$

Osmometry

$$\Pi/[J] \approx RT + BRT[J]$$



$$\frac{\Pi}{c_{\text{mass},J}} \approx \frac{RT}{M} + \frac{BRT}{M^2} c_{\text{mass},J}$$



Summary

- Phase
- Chemical potential
- Gibbs-Duhem eq.



Multi-component
system

Gas mixture

- Dalton's law
- Perfect gas
- Real gas

Liquid mixture

Ideal solution

- Raoult's law
$$p_A = x_A p_A^*$$
- Chemical potential
$$\mu_A = \mu_A^* + RT \ln x_A$$
- $\Delta_{\text{mix}}G = -T\Delta_{\text{mix}}S$

Real solution

- Regular solution
- Bragg-Williams model
- Excess function
- Activity

$$\mu_A = \mu_A^* + RT \ln a_A$$
$$\mu_B = \mu_B^\ominus + RT \ln a_B$$

Ideal-dilute solution

- Henry's law
$$p_B = x_B K_B$$
- Chemical potential
$$\mu_B = \mu_B^\ominus + RT \ln x_B$$

- L-G phase diagram
- L-L phase diagram
- L-S phase diagram

Colligative properties

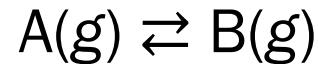
8. Chemical Equilibria

Chapter Overview

- Chemical reactions and equilibria
 - Reaction quotient and equilibrium constant
 - Activities
- Le Chatelier's principle
 - Amount change
 - Volume change
 - Temperature change



Simple Reaction



- Extent of reaction ξ : for $d\xi$, $dn_A = -d\xi$ and $dn_B = +d\xi$

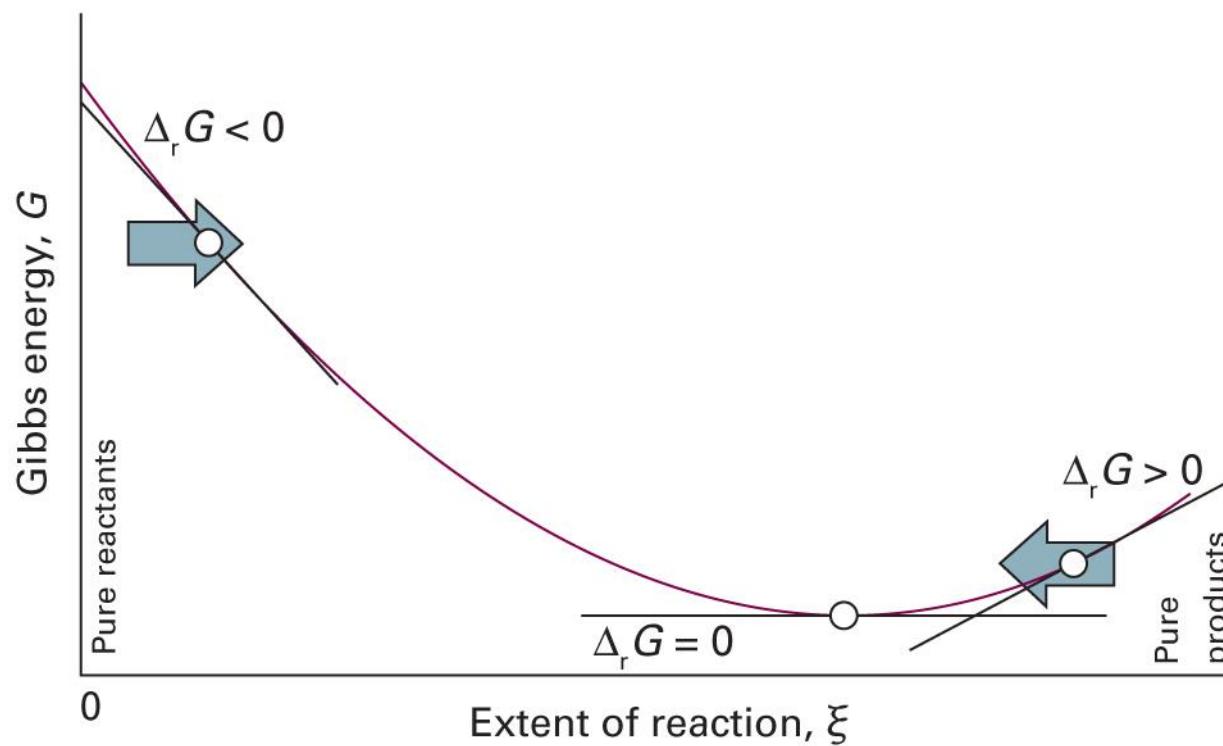
$$dG = \mu_A dn_A + \mu_B dn_B = -\mu_A d\xi + \mu_B d\xi = (\mu_B - \mu_A) d\xi$$

- Reaction Gibbs energy

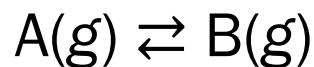
$$\Delta_r G = \left(\frac{\partial G}{\partial \xi} \right)_{p,T} = \mu_B - \mu_A$$

Progress of Reaction

$$\Delta_r G = \mu_B - \mu_A$$



Reaction Quotient



For a perfect gas, $\mu = \mu^\ominus + RT \ln(p/p^\ominus)$

$$\Delta_r G = \mu_B - \mu_A = \left\{ \mu_B^\ominus + RT \ln \left(\frac{p_B}{p_B^\ominus} \right) \right\} - \left\{ \mu_A^\ominus + RT \ln \left(\frac{p_A}{p_A^\ominus} \right) \right\}$$

Hence, using $\Delta_r G^\ominus = \mu_B^\ominus - \mu_A^\ominus$,

$$\Delta_r G = \Delta_r G^\ominus + RT \ln \left(\frac{p_B}{p_A} \right) = \Delta_r G^\ominus + RT \ln Q$$

Equilibrium Constant

$$\Delta_r G = \Delta_r G^\ominus + RT \ln Q$$

At equilibrium, $\Delta_r G = 0$

$$0 = \Delta_r G^\ominus + RT \ln Q_{\text{eq}} = \Delta_r G^\ominus + RT \ln K$$

Hence,

$$\Delta_r G^\ominus = -RT \ln K, \text{ where } K = (p_{\text{B}}/p_{\text{A}})_{\text{eq}}$$

Generalization

For an arbitrary reaction,

$$\sum_{\text{reactant } i} \nu(i)i \rightleftharpoons \sum_{\text{product } j} \nu(j)j$$



$$\nu_{\text{A}} = 2, \nu_{\text{B}} = 1, \nu_{\text{C}} = 3, \nu_{\text{D}} = 1$$

Now,

$$dG = \sum_{\text{product } j} \mu_j dn_j + \sum_{\text{reactant } i} \mu_i dn_i$$

Generalization

$$dG = \sum_{\text{product } j} \mu_j dn_j + \sum_{\text{reactant } i} \mu_i dn_i$$

$$dG = \sum_{\text{product } j} \mu_j \nu_j d\xi - \sum_{\text{reactant } i} \mu_i \nu_i d\xi$$

$$dG = \left(\sum_{\text{product } j} \mu_j \nu_j - \sum_{\text{reactant } i} \mu_i \nu_i \right) d\xi$$

$$\Delta_r G = \left(\frac{\partial G}{\partial \xi} \right)_{p,T} = \sum_{\text{product } j} \mu_j \nu_j - \sum_{\text{reactant } i} \mu_i \nu_i$$

Chemical Potential and Activity

- Real gas: $\mu = \mu^\ominus + RT \ln(f/p^\ominus)$
- Real solution (solvent): $\mu = \mu^* + RT \ln a$
- Real solution (solute): $\mu = \mu^\ominus + RT \ln a$
- General: $\mu = \mu^\ominus + RT \ln a$

Component	Basis	Standard state	Activity	Limits
Solid or liquid		Pure, 1 bar	$a = 1$	
Solvent	Raoult	Pure solvent, 1 bar	$a = p/p^*, a = \gamma x$	$\gamma \rightarrow 1$ as $x \rightarrow 1$ (pure solvent)
Solute	Henry	(1) A hypothetical state of the pure solute	$a = p/K, a = \gamma x$	$\gamma \rightarrow 1$ as $x \rightarrow 0$
		(2) A hypothetical state of the solute at molality $b^\ominus$	$a = \gamma b/b^\ominus$	$\gamma \rightarrow 1$ as $b \rightarrow 0$
Gas	Fugacity [†]	Pure, a hypothetical state of 1 bar and behaving as a perfect gas	$f = \gamma p$	$\gamma \rightarrow 1$ as $p \rightarrow 0$



Generalization

$$\Delta_r G = \sum_{\text{product } j} \nu_j \mu_j - \sum_{\text{reactant } i} \nu_i \mu_i$$

For each species x ,

$$\nu_x \mu_x = \nu_x (\mu_x^\ominus + RT \ln a_x) = \nu_x \mu_x^\ominus + RT \ln(a_x^{\nu_x})$$

Then

$$\begin{aligned} \sum_x \nu_x \mu_x &= \sum_x \nu_x \mu_x^\ominus + RT \sum_x \ln(a_x^{\nu_x}) \\ &= \sum_x \nu_x \mu_x^\ominus + RT \ln \prod_x a_x^{\nu_x} \end{aligned}$$

Generalization

$$\Delta_r G = \sum_{\text{product } j} \nu_j \mu_j - \sum_{\text{reactant } i} \nu_i \mu_i$$

Substitution leads to

$$\Delta_r G = \Delta_r G^\ominus + RT \ln \prod_{\text{product } j} a_j^{\nu_j} - RT \ln \prod_{\text{reactant } i} a_i^{\nu_i}$$

where $\Delta_r G^\ominus = \sum_{\text{product } j} \nu_j \mu_j^\ominus - \sum_{\text{reactant } i} \nu_i \mu_i^\ominus$

$$\Delta_r G = \Delta_r G^\ominus + RT \ln \left(\prod_{\text{product } j} a_j^{\nu_j} / \prod_{\text{reactant } i} a_i^{\nu_i} \right)$$

Q and K

$$\Delta_r G = \Delta_r G^\ominus + RT \ln Q$$

where the **reaction quotient** Q is defined as

$$Q = \frac{\prod_{\text{product } j} a_j^{v_j}}{\prod_{\text{reactant } i} a_i^{v_i}}$$

At equilibrium, $\Delta_r G = 0$

$$0 = \Delta_r G^\ominus + RT \ln Q_{\text{eq}} = \Delta_r G^\ominus + RT \ln K$$

where the **equilibrium constant** K is Q at equilibrium